

Lung adenocarcinoma: preserve the finding, remove the identifiers

A detailed synthetic biopsy and linked EGFR report, processed by unchanged local methods.

Original synthetic report
NEW SYNTHETIC ILLUSTRATION FROZEN LOCAL METHODS OUTSIDE BOTH BENCHMARK COHORTS SYNTHETIC PATHOLOGY LABORATORY ILLUSTRATIVE CASE - NO REAL PATIENT Patient name: Lena Varnell Medical record number: 48271639 Date of birth: 06/18/1965 Ordering provider: Mira Norvale, MD SURGICAL PATHOLOGY REPORT Accession: SP32-041728 Collected: 09/10/2032 09:25 Received: 09/10/2032 10:12 Reported: 09/12/2032 15:40 SPECIMEN A. Right upper lobe lung, image-guided core biopsy FINAL DIAGNOSIS Lung, right upper lobe, core biopsy: Pulmonary adenocarcinoma. MICROSCOPIC DESCRIPTION Core fragments contain infiltrating malignant glands with focal papillary architecture, nuclear enlargement, and conspicuous nucleoli. Focal necrosis is present. No definitive squamous differentiation is identified in the sampled tissue. IMMUNOHISTOCHEMISTRY TTF-1: Positive in tumor nuclei. Napsin A: Positive in tumor cytoplasm. p40: Negative in tumor cells. The findings support pulmonary adenocarcinoma. CLINICAL HISTORY Known EGFR-mutated lung adenocarcinoma. Repeat biopsy for molecular reassessment; correlate with prior therapy and testing. GROSS DESCRIPTION Four tan-white cores, 0.3-0.9 cm in length and up to 0.1 cm in diameter; submitted entirely in cassette A1. Fixative: 10% neutral buffered formalin. MOLECULAR TESTING Block A1 selected; estimated viable tumor cellularity 40%. Targeted DNA sequencing reported separately as MP32-006381. The diagnosis is based on this biopsy; pathologic stage is not assigned. Electronically signed: Arlen Voss, MD Lena Varnell 48271639 SP32-041728 Page 1 of 1 Exact source text or saved redaction; highlighting is a display overlay. All identities and assay metrics are invented.

Original synthetic report

NEW SYNTHETIC ILLUSTRATION | FROZEN LOCAL METHODS | OUTSIDE BOTH BENCHMARK COHORTS

SYNTHETIC PATHOLOGY LABORATORY

ILLUSTRATIVE CASE - NO REAL PATIENT

Patient name: Lena Varnell

Medical record number: 48271639

Date of birth: 06/18/1965

Ordering provider: Mira Norvale, MD

MOLECULAR PATHOLOGY REPORT

Accession: MP32-006381

Parent accession: SP32-041728

Collected: 09/10/2032 09:25

Received: 09/13/2032 11:05

Reported: 09/18/2032 14:20

SPECIMEN INFORMATION

Site: Right upper lobe lung

Diagnosis: Pulmonary adenocarcinoma

Material: Formalin-fixed paraffin-embedded core biopsy

Specimen part: A | Block: A1

Tumor cellularity: 40% | Macrodissection: Performed

MOLECULAR RESULTS

EGFR p.L858R detected; variant allele fraction 28%.

EGFR p.T790M detected; variant allele fraction 12%.

VARIANT DETAILS

Gene	Transcript	Exon	Coding change	Protein change
EGFR	NM_005228.5	21	c.2573T>G	p.L858R
EGFR	NM_005228.5	20	c.2369C>T	p.T790M

INTERPRETATION

Two EGFR substitutions are detected. Correlate with prior molecular testing and treatment history. The phase relationship of the two variants is not determined. Tumor-only testing does not establish whether a variant is somatic or germline.

QUALITY METRICS

DNA input: 42 ng | Mean target depth: 1540x

L858R locus depth: 1640x | T790M locus depth: 1240x

Target bases at >=250x coverage: 99.4%

All listed quality metrics are simulated for this illustration.

METHOD

Illustrative targeted DNA sequencing of EGFR exons 18-21.

Single-nucleotide variants and small insertions/deletions assessed. Illustrative reporting threshold: 5% variant allele fraction.

No RNA fusion or copy-number assessment is included.

LIMITATIONS

Low tumor content, sampling, and variants below the reporting threshold may limit detection. A negative result would not exclude alterations outside the interrogated regions. No paired normal sample was analyzed. Assay specifications are illustrative only.

Electronically signed: Arlen Voss, MD

Lena Varnell | 48271639 | MP32-006381 | Page 1 of 1

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MOLECULAR RESULT - EXACT EXCERPTS

Original synthetic report

EGFR p.T790M detected

New illustrative case; excluded from both 200-pair experiments. No hosted LLM was run on these reports. The variant is information to retain, not the identifying target. Names and record numbers are identifiers. Full-size JPEGs and raw text are available online.

Institution-developed pipeline
NEW SYNTHETIC ILLUSTRATION FROZEN LOCAL METHODS OUTSIDE BOTH BENCHMARK COHORTS SYNTHETIC PATHOLOGY LABORATORY ILLUSTRATIVE CASE - ** **** * Patient name: **** * ***** ***** number: ***** Date of birth: **/**/**** Ordering provider: **** *, MD SURGICAL PATHOLOGY REPORT Accession: ****-***** Collected: **/**/**** 09:25 Received: **/**/**** 10:12 Reported: **/**/**** 15:40 SPECIMEN A. Right upper lobe lung, image-guided core biopsy FINAL DIAGNOSIS Lung, right upper lobe, core biopsy: Pulmonary adenocarcinoma. MICROSCOPIC DESCRIPTION Core fragments contain infiltrating malignant glands with focal papillary architecture, nuclear enlargement, and conspicuous nucleoli. Focal necrosis is present. No definitive squamous differentiation is identified in the sampled tissue. IMMUNOHISTOCHEMISTRY TTF-1: Positive in tumor nuclei. ***** A: Positive in tumor cytoplasm. ***: Negative in tumor cells. The findings support pulmonary adenocarcinoma. CLINICAL HISTORY Known EGFR-mutated lung adenocarcinoma. Repeat biopsy for molecular reassessment; correlate with prior therapy and testing. GROSS DESCRIPTION Four tan-white cores, 0.3-0.9 cm in length and up to 0.1 cm in diameter; submitted entirely in cassette **. Fixative: 10% neutral buffered formalin. MOLECULAR TESTING Block A1 selected; estimated viable tumor cellularity 40%. Targeted DNA sequencing reported separately as ****-*****. The diagnosis is based on this biopsy; pathologic stage is not assigned. Electronically signed: **** *, MD **** * ***** ****-***** Page 1 of 1 Exact source text or saved redaction; highlighting is a display overlay. All identities and assay metrics are invented.

Institution-developed pipeline

NEW SYNTHETIC ILLUSTRATION | FROZEN LOCAL METHODS | OUTSIDE BOTH BENCHMARK COHORTS

SYNTHETIC PATHOLOGY LABORATORY

ILLUSTRATIVE CASE - ** **** *

Patient name: **** *

***** ***** number: *****

Date of birth: **/**/****

Ordering provider: **** *, MD

MOLECULAR PATHOLOGY REPORT

Accession: ****-*****

Parent accession: ****-*****

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EGFR p.**** detected; variant allele fraction 12%.

VARIANT DETAILS

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EGFR	NM_****.5	20	c.*****>T	p.****

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Two EGFR substitutions are detected. Correlate with prior molecular testing and treatment history. The phase relationship of the two variants is not determined. Tumor-only testing does not establish whether a variant is somatic or germline.

QUALITY METRICS

DNA input: 42 ng | Mean target depth: ****

**** locus depth: **** | **** locus depth: ****

Target bases at >=250x coverage: **.4%

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**** * | ***** | ****-***** | Page 1 of 1

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Institution-developed pipeline

EGFR p.***** detected

Refined local rules
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Refined local rules

NEW SYNTHETIC ILLUSTRATION | FROZEN LOCAL METHODS | OUTSIDE BOTH BENCHMARK COHORTS

SYNTHETIC PATHOLOGY LABORATORY

ILLUSTRATIVE CASE - ** **** *

Patient name: *****

***** ***** number: *****

Date of birth: *****

Ordering provider: *****

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